

"We are what we repeatedly do; excellence, then, is not an act but a habit."

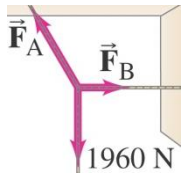
- Aristotle (384 BC – 322 BC)

Outline

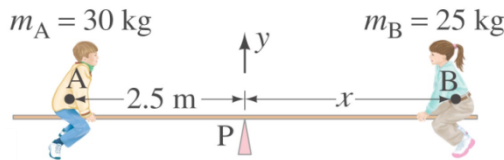
Conditions for Equilibrium

Translational Equilibrium:

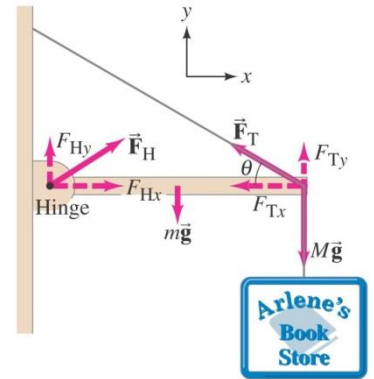
$$\sum \vec{F} = 0$$



Example of Systems in Equilibrium



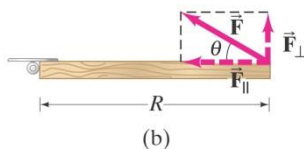
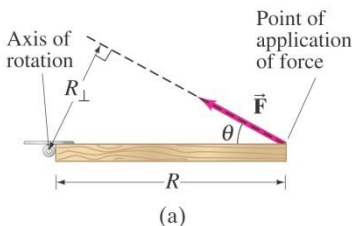
Giancoli pg 314 Fig 12.6:
See-saw Physics



Giancoli pg316 Fig 12.9

Rotational Equilibrium:

$$\sum \vec{\tau} = \sum R F \sin \theta = 0$$



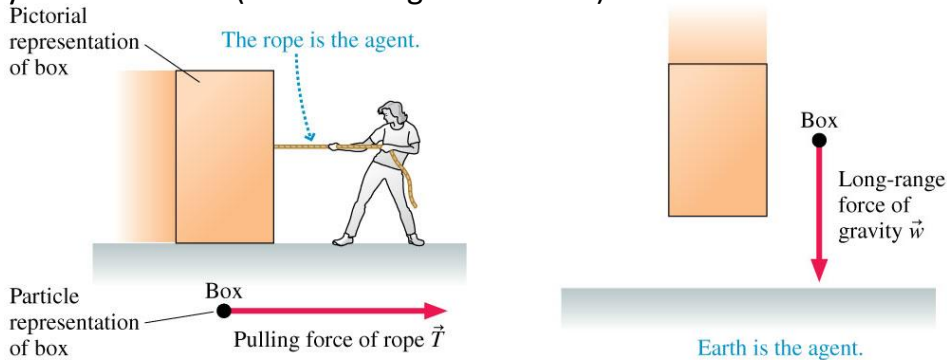
Giancoli Fig 10.14: Torque

Self-check – after completing the lessons (lectures and tutorials) for this topic, are you able to

1. solve problems involving forces acting on extended object either in equilibrium or experiencing angular acceleration
 2. determine the center of mass of a given object
 3. determine the moment of inertia of a given object
- solve problems involving rotating objects (such as pulleys or hinged sticks) in a system using either conservation of energy or Newton's laws of motion

Point Object -> Extended Object

So far, when dealing with forces, we have simplified our analysis by treating the objects of interest as point objects as shown below. These point objects only move in a line (can be straight or curved).



So typically, we can apply Newton's second law of motion to analyze forces on a point object:

$$\sum \vec{F} = m\vec{a}.$$

When the system is in equilibrium, $\sum \vec{F} = m\vec{a} = 0$.

However, in reality, the point at which the force is exerted on an object is also very important as this may cause the object to rotate. Here are some examples. In these cases, we need to consider where the forces are acting and how the mass is distributed in the object.



For extended objects, we need to think about where the force is acting with respect to a reference point to compute the turning effect given by **torque τ** . The equivalent to Newton's law is

$$\sum \vec{\tau} = I\vec{\alpha}$$

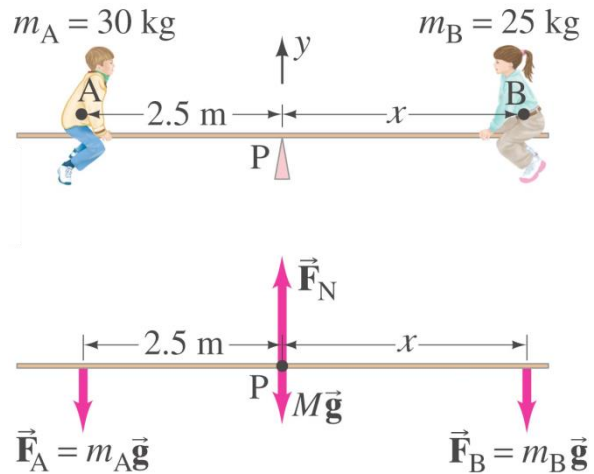
<https://www.youtube.com/watch?v=GgNcQx3uaXc&t=9s>



And I gives the rotational inertia and α , the angular acceleration. This may all sound too difficult so we will start with simple examples where $\sum \vec{\tau} = 0$.

Torque: Giancoli pg 314 Example 12-3 (seesaw)

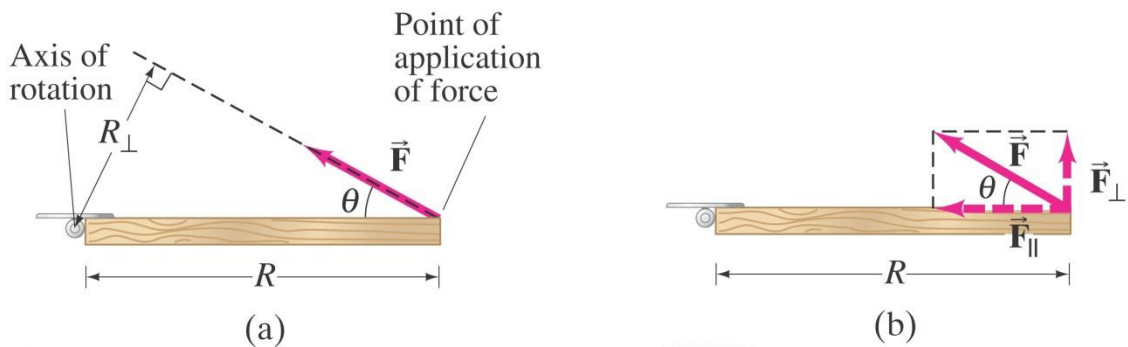
A board of mass $M = 2.0 \text{ kg}$ serves as a seesaw for two children. Child A has a mass of 30 kg and sits 2.5 m from the pivot point, P (his center of gravity is 2.5 m from the pivot). At what distance x from the pivot must child B, of mass 25 kg , place herself to balance the seesaw? Assume the board is uniform and centered over the pivot.



Giancoli pg 314 Fig 12.6:
See-saw Physics

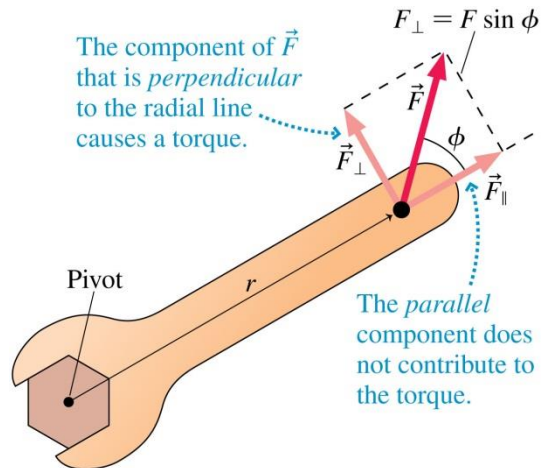
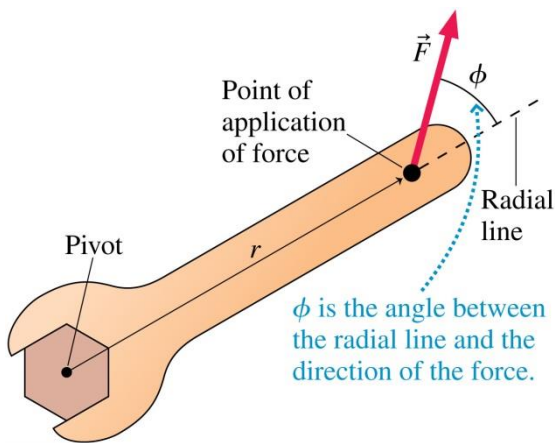
Definition of Torque: Giancoli pg 256-7

Torque is the product of force with the perpendicular distance from the axis of rotation to the line which the force acts (lever arm / moment arm).

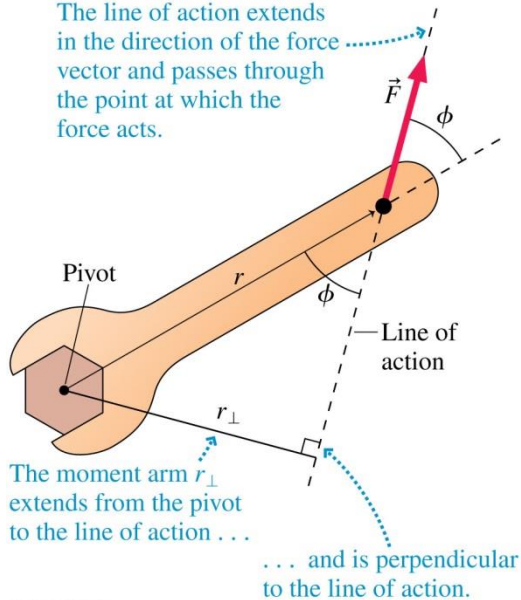


Giancoli pg 257 Fig 10.14:
Two perspectives of the definition of torque

Example on torque: Knight pg 233



The line of action extends in the direction of the force vector and passes through the point at which the force acts.



Knight Fig 7.17-7.19: Torque due to force acting on a wrench

Knight pg 235 Ex 7.10

Luis uses a 20 cm wrench to tighten a nut, turning it clockwise. The wrench handle is tilted 30° above the horizontal, and Luis pulls straight down on the end with a force of 100 N. How much torque does Luis exert on the nut?

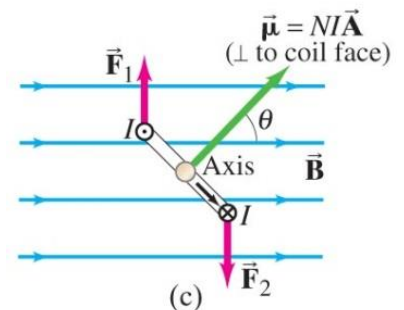
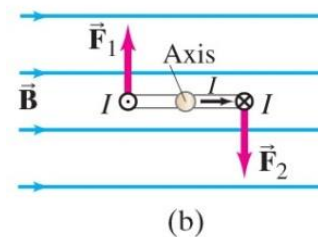
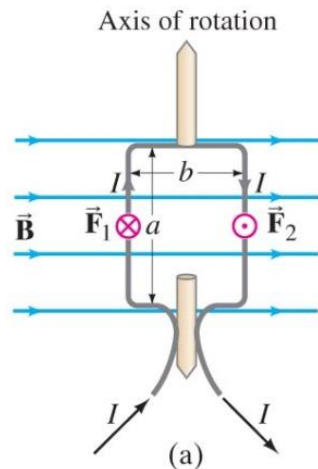
Giancoli pg 719 Example 27.11 (modified)

A rectangular coil of wire with $b = 3\text{ cm}$ and $a = 5\text{ cm}$ and contains 10 loops. The current in each loop is 3.00 A , and the coil is placed in a 2.00-T external magnetic field. Determine the maximum and minimum torque exerted on the coil by the field.

Force on wire, $|F_1| = |F_2| =$

Maximum torque $\tau_{max} =$

Minimum torque $\tau_{min} =$



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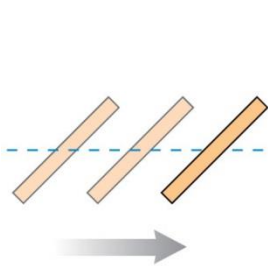
Conditions for Equilibrium

Translational Equilibrium

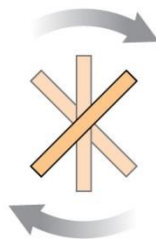
The vector sum of forces acting on an object is zero.

Rotational Equilibrium

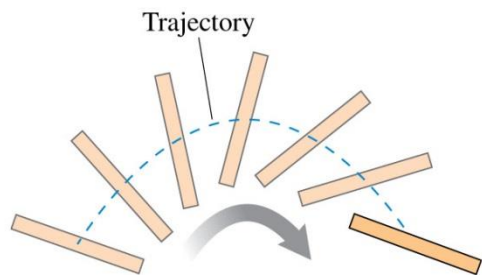
The net torque on the object about any pivot is zero.



Translational motion:
The object as a whole
moves along a trajectory
but does not rotate.



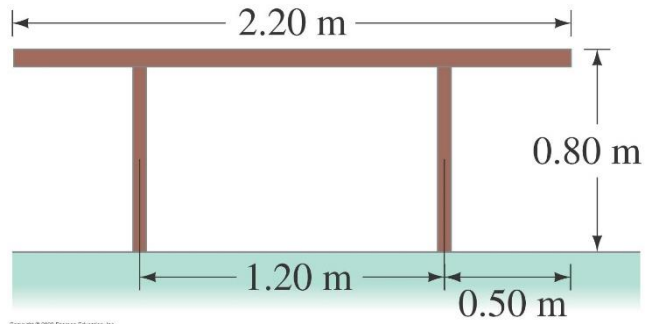
Rotational motion:
The object rotates about a
fixed point. Every point on
the object moves in a circle.



Combination motion:
An object rotates as it
moves along a trajectory.

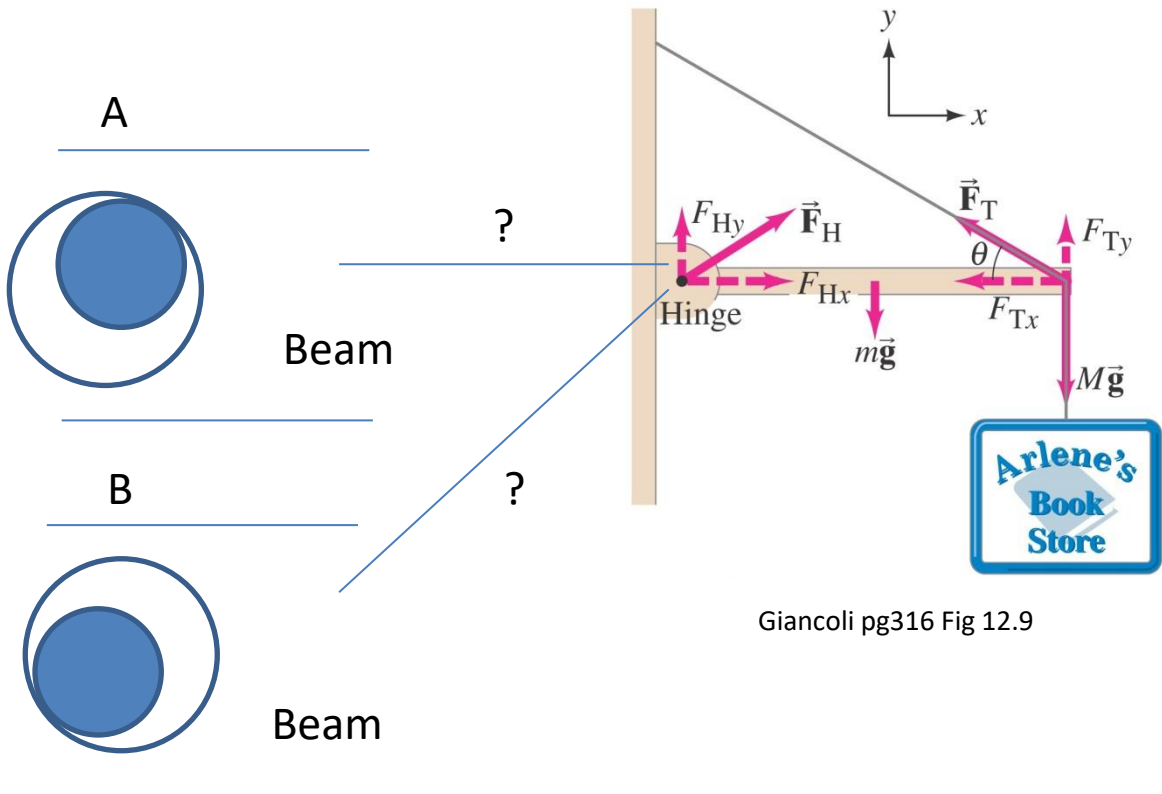
More Examples

Giancoli P12-13: How close to the edge of the 24.0-kg table shown in the figure can a 66.0-kg person sit without tipping it over?



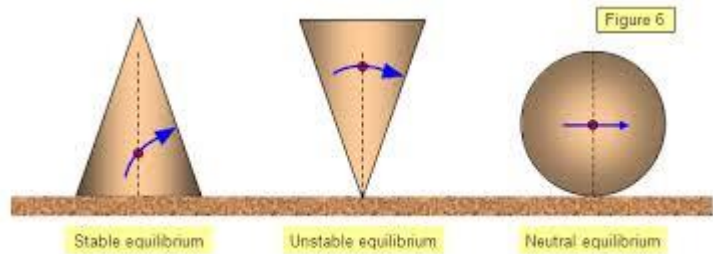
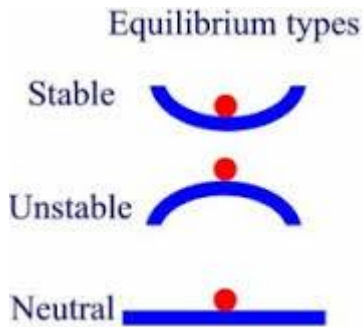
Example on Equilibrium: Giancoli pg 316 Example 12-5

A uniform beam, 2.20 m long with mass $m = 25.0$ kg, is mounted by a small hinge on a wall. The beam is held in a horizontal position by a cable that makes an angle $\theta = 30.0^\circ$. The beam supports a sign of mass $M = 28.0$ kg suspended from its end. Determine the components of the force $\vec{F}_H$ that the (smooth) hinge exerts on the beam, and the tension $\vec{F}_T$ in the supporting cable.



Giancoli pg316 Fig 12.9

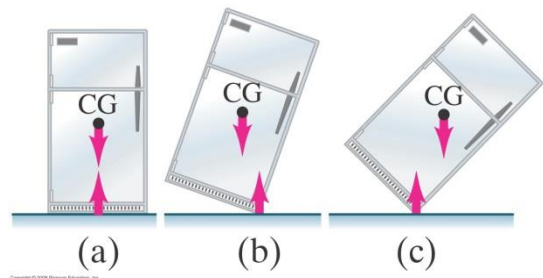
Equilibrium and Stability



<https://serc.carleton.edu/introgeo/models/EquStBOT.html>

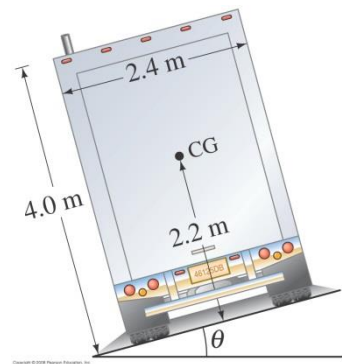
http://www.schoolphysics.co.uk/age11-14/Mechanics/Statics/text/Stability_/index.html

An object in stable equilibrium may become unstable if it is tipped so that its center of gravity is outside the pivot point. Of course, it will be stable again once it lands!



Giancoli pg 318 Fig 12.12

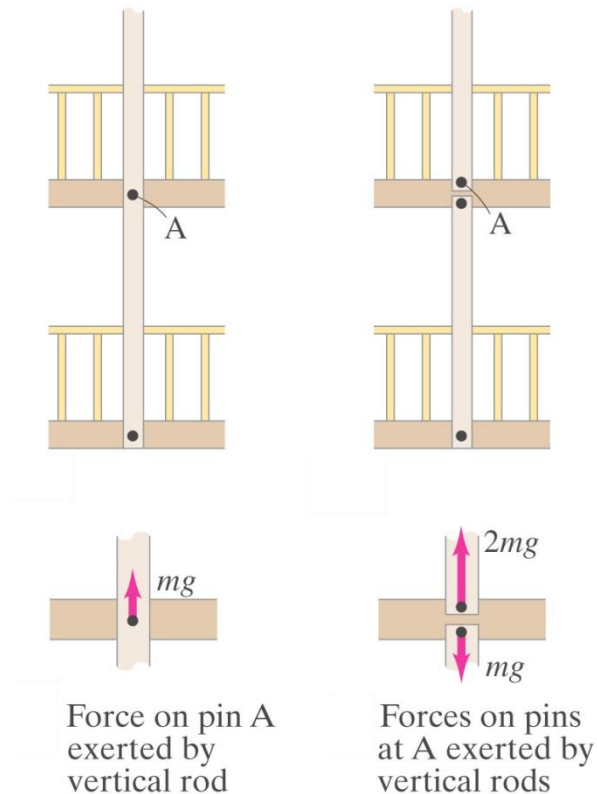
[G12.63] The center of gravity of a loaded truck depends on how the truck is packed. If it is 4.0 m high and 2.4 m wide, and its CG is 2.2 m above the ground, how steep a slope can the truck be parked on without tipping over?



Engineering Applications

Giancoli Conceptual Example 12-9: A tragic substitution.

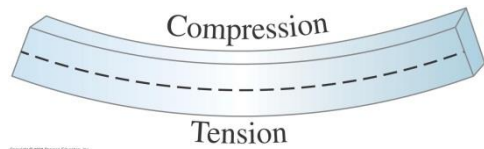
Two walkways, one above the other, are suspended from vertical rods attached to the ceiling of a high hotel lobby. The original design called for single rods 14 m long, but when such long rods proved to be unwieldy to install, it was decided to replace each long rod with two shorter ones as shown. Determine the net force exerted by the rods on the supporting pin A (assumed to be the same size) for each design. Assume each vertical rod supports a mass m of each bridge.



In the first design, the upward force on pin A is mg , which is the downward force exerted on it by the upper bridge. In the second design, the lower pin still has a force (now downward) on it of mg , but the upper pin, which is supporting both bridges, has a force of $2mg$. This bridge collapsed after it was put into use, resulting in the loss of more than 100 lives.

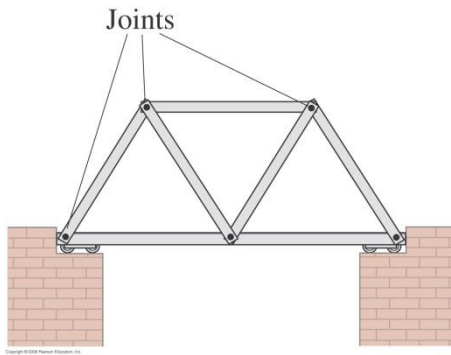
Engineering Applications: Trusses and Bridges (Giancoli pg 324-5)

A beam sags, either under its own weight or when loaded



Giancoli pg 322 Fig 12.21:
Bending a beam

How can we support a long span of beam so that it doesn't sag?

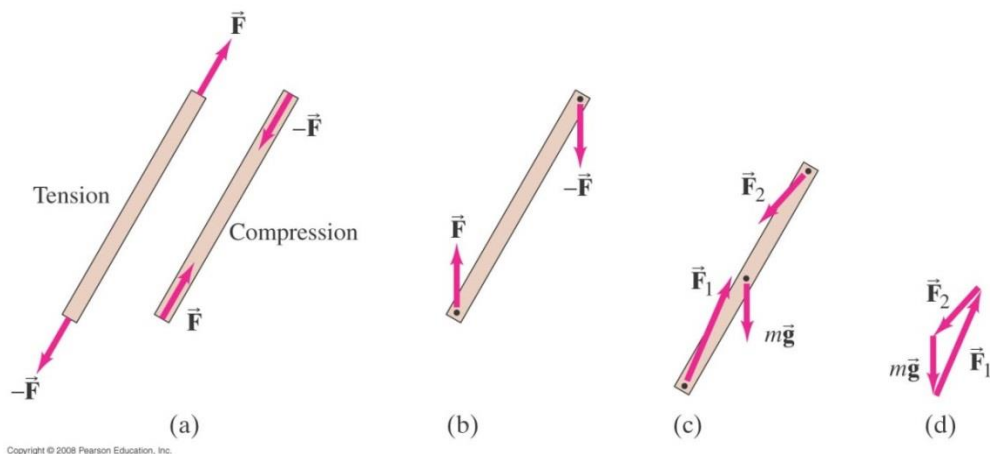


Giancoli pg 324 Fig 12.25:
A truss bridge



Giancoli pg 324 Fig 12.26:
A roof truss

A **truss** is a framework of rods or **struts** joined together (usually in triangles) at their ends by pins or rivets. The place where the struts are joined by a pin is known as the **joint**. This is one of the ways to span wide spaces. Each truss member is under either tension or compression; if the mass is small, these forces act along the strut.



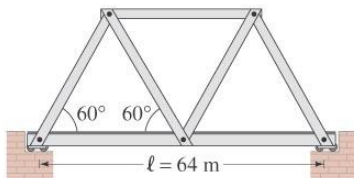
Giancoli pg 324 Fig 12.27
Forces acting on a strut

Engineering Applications: Trusses and Bridges (Giancoli pg 324-5)

MAE, CE – Video after Exam

Example 12-11: A truss bridge.

Determine the tension or compression in each of the struts of the truss bridge shown. The bridge is 64 m long and supports a uniform level concrete roadway whose total mass is 1.40×10^6 kg. Use the method of joints, which involves (1) drawing a free-body diagram of the truss as a whole, and (2) drawing a free-body diagram for each of the pins (joints), one by one, and setting $\Sigma \vec{F} = 0$ for each pin. Ignore the mass of the struts. Assume all triangles are equilateral.

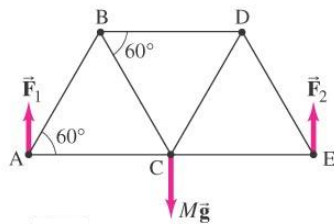


From symmetry, we see that each end supports half the weight of the bridge:

$$F_1 = F_2 = \frac{Mg}{2}$$

$\vec{F}_{AB}$ means the force exerted by the strut AB.

$\vec{F}_{AC}$ means the force exerted by the strut AC.



At pin A,

$$\Sigma F_x = F_{AC} - F_{AB} \cos 60^\circ = 0$$

$$\Sigma F_y = F_1 - F_{AB} \sin 60^\circ = 0$$

Thus,

$$F_{AB} = \frac{F_1}{\sin 60^\circ} = \frac{\frac{1}{2}Mg}{\frac{1}{2}\sqrt{3}} = \frac{1}{\sqrt{3}}Mg$$

And

$$F_{AC} = F_{AB} \cos 60^\circ = \frac{1}{2\sqrt{3}}Mg$$

At Pin B,

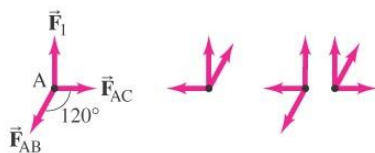
$$\Sigma F_x = F_{BA} \cos 60^\circ + F_{BC} \cos 60^\circ - F_{BD} = 0$$

$$\Sigma F_y = F_{BA} - F_{BC} \sin 60^\circ = 0$$

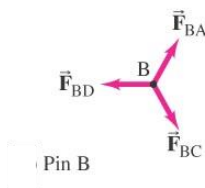
Thus, $F_{BA} = F_{AB}$, we have

$$F_{BC} = F_{AB} = \frac{1}{\sqrt{3}}Mg$$

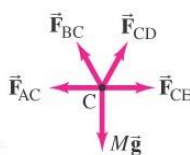
$$\text{And } F_{BD} = F_{AB} \cos 60^\circ + F_{BC} \cos 60^\circ = \frac{1}{\sqrt{3}}Mg.$$



Pin A (different guesses)



Pin B



Pin C

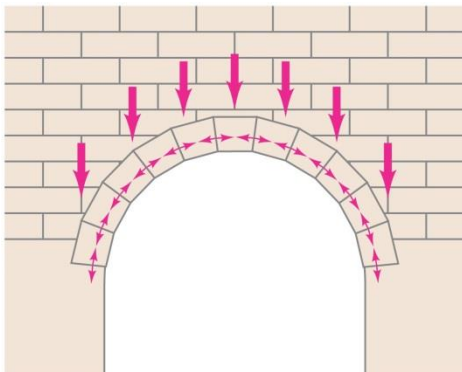
Engineering Applications: Trusses and Bridges



<https://www.youtube.com/watch?v=VOnRPefcno&t=4s>



Giancoli Figure 12.30



Giancoli Figure 12.34



<http://www.targus.com/us/truss-case-for-hp-touchpad-thz07202us>



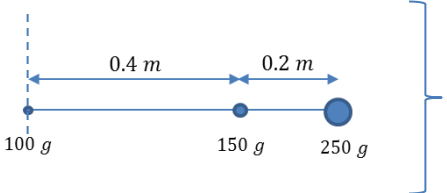
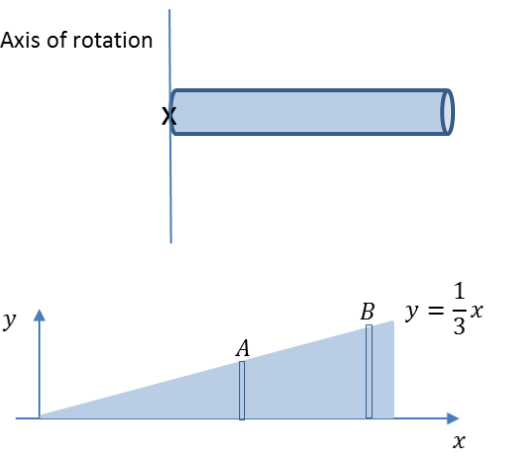
2026S1 PH1012: Physics A
System of Masses / Torque / Circular Motion (Recap)

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Week 12
Serway Chap 9.6 – 9.7

Obstacles are those frightful things you see when you take your eyes off your goal.

- Henry Ford (1863-1947)

Discrete 	Total Mass $M = \sum_i m_i$	Center of Mass $x_{CM} = \frac{\sum_i m_i x_i}{M}$	Moment of Inertia (about an axis) $I = \sum_i m_i x_i^2$
Continuous 	$M = \int dm$	$x_{CM} = \frac{1}{M} \int x dm$	$I = \int x^2 dm$

Center of Mass (Gravity)

Consider the earlier question that we dealt with to find where girl should sit in order for the see-saw to be balanced. Recall that we used net torque = 0 for rotational equilibrium and net force = 0 for translational equilibrium.

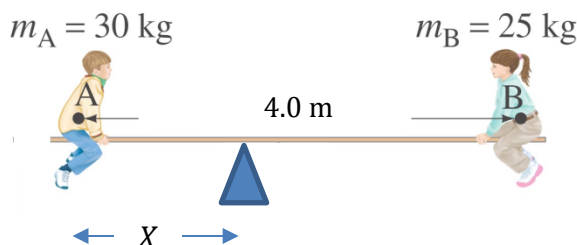
Here, we will assume that the see-saw plank is very light (massless) to keep our discussion simple.

Taking moments about the pivot P , we can write

$$\sum \tau = 30 \times 9.8 \times 2.5 - 25 \times 9.8 \times x = 0$$

and solve for x and we see that $x = 3.0 \text{ m}$. Now consider another problem:

The same two children are sitting on the (massless) see-saw. The two children are fixed at 4.0 m apart. Where should the pivot be placed (from the boy) in order for the see-saw to balance?



Quite naturally, we will write

$$\sum \tau = 30 \times 9.8 \times X - 25 \times 9.8 \times (4.0 - X) = 0 \text{ and simplifying we get}$$

$$(30 + 25) \times 9.8 \times X = 25 \times 9.8 \times 4.0$$

We notice two things here:

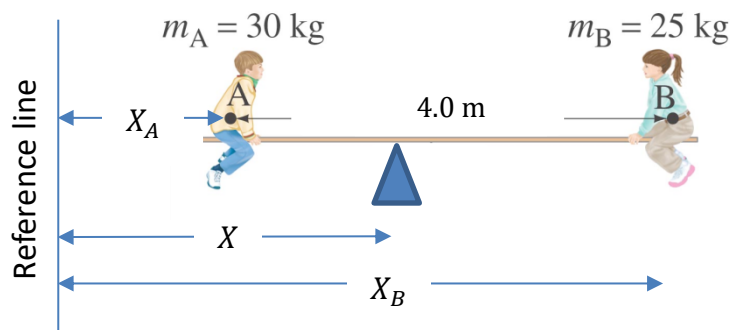
- i. We are using the boy as a reference for measuring X . How can this method be generalized so that we can handle a scenario where there are more people sitting on the see-saw.
- ii. All the $g = 9.8 \text{ m/s}^2$ can be cancelled.

To be very general, we can in fact write

$$(M_A + M_B) \times X_{CM} = M_B \times X_B + M_A \times X_A$$

Where X_A and X_B are the distances to the reference point. So here $X_A = 0$. X_{CM} is the distance from the reference point to the center of mass.

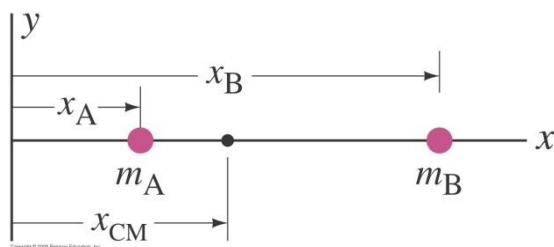
Center of Mass (Gravity)



We see that the reference is arbitrary, just that we have to measure all the distances of the mass to it.

The **center of mass** (CM) is the unique point at the center of a distribution of mass in space that has the property that the weighted position vectors relative to this point sum to zero. We see that if the size of mass distribution is small such that g is the same throughout the object, the center of mass will coincide with the center of gravity (CG) – the point where the resultant torque of a system of mass vanishes – i.e. if you put a pivot at the CG, the system of mass will balance on it.

For a system of two masses, m_A and m_B with coordinates as shown below



We can write

$$(m_A + m_B) x_{CM} = m_A x_A + m_B x_B$$

The location of the center of mass x_{CM} is thus

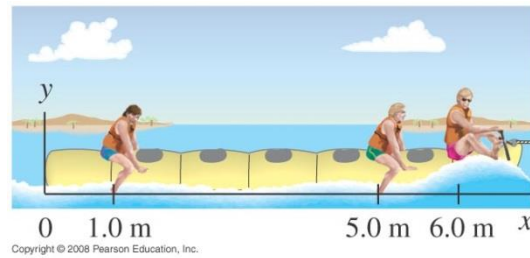
$$x_{CM} = \frac{m_A x_A + m_B x_B}{m_A + m_B} = \frac{m_A x_A + m_B x_B}{M},$$

where M is the total mass of the system. We note here that if $m_A > m_B$, x_{CM} will be closer to m_A . We make a generalization to a system of n particles along a straight line – the CM of the system is located at

$$x_{CM} = \frac{m_1 x_1 + m_2 x_2 + \cdots + m_n x_n}{m_1 + m_2 + \cdots + m_n} = \frac{\sum_{i=1}^n m_i x_i}{M}$$

Example 9-14: CM of three guys on a raft.

Three people of roughly equal masses m on a lightweight (air-filled) banana boat sit along the x axis at positions $x_A = 1.0$ m, $x_B = 5.0$ m, and $x_C = 6.0$ m, measured from the left-hand end. Find the position of the CM. Ignore the boat's mass.



Consider if there is a class test and there are three students, their scores are 1 point, 5 points and 6 points respectively, what is the average score of the three?

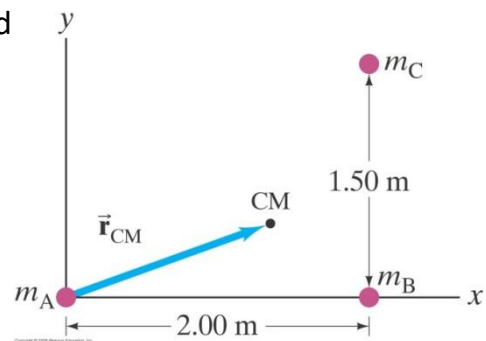
The coordinates of the CM of an extended object in two or three dimensions is given by

$$x_{CM} = \frac{\sum_{i=1}^n m_i x_i}{M}; y_{CM} = \frac{\sum_{i=1}^n m_i y_i}{M}; z_{CM} = \frac{\sum_{i=1}^n m_i z_i}{M}.$$

Giancoli pg 231 Example 9-15: Three particles in 2-D.

Three particles, each of mass 2.50 kg, are located at the corners of a right triangle whose sides are 2.00 m and 1.50 m long, as shown.

Locate the center of mass.



Giancoli pg 231 Fig 9.25:
Finding CM of three particles

$$x_{CM} = \frac{((2.50 \text{ kg})(0 \text{ m}) + (2.50 \text{ kg})(2.00 \text{ m}) + (2.50 \text{ kg})(2.00 \text{ m}))}{3(2.50 \text{ kg})} = 1.33 \text{ m}$$
$$y_{CM} = \frac{((2.50 \text{ kg})(0 \text{ m}) + (2.50 \text{ kg})(0 \text{ m}) + (2.50 \text{ kg})(1.50 \text{ m}))}{3(2.50 \text{ kg})} = 0.50 \text{ m}$$

Center of Mass of extended object

Example 9-16(a): CM of a thin rod.

Show that the CM of a uniform thin rod of length l and mass M is at its center. Let the mass per unit length be $\lambda = M/l$.

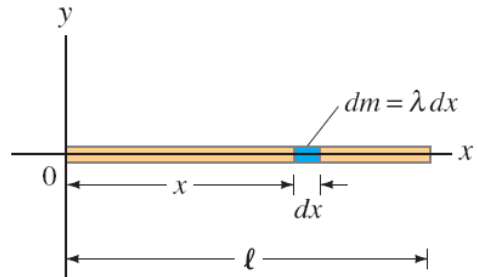
So instead of

$$x_{CM} = \frac{\sum_i m_i x_i}{M}$$

We compute

$$x_{CM} = \frac{\int x \, dm}{M}$$

$$x_{CM} = \frac{1}{M} \int_{x=0}^l x \, dm =$$

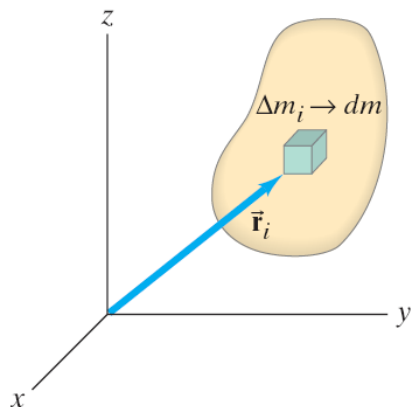


For an extended object, we imagine making it up of tiny particles, each of tiny mass, and adding up the product of each particle's mass with its position and dividing by the total mass. In the limit that the particles become infinitely small, this gives:

$$x_{CM} = \frac{\int x \, dm}{M};$$

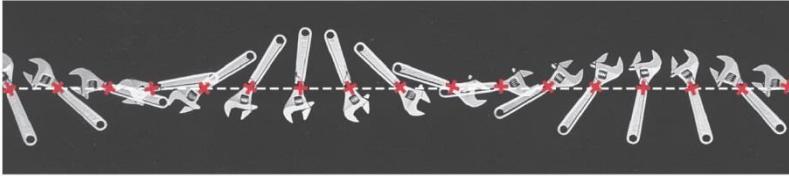
$$y_{CM} = \frac{\int y \, dm}{M};$$

$$z_{CM} = \frac{\int z \, dm}{M};$$



Center of Mass and Motion of Extended Objects

Translational motion of a (rotating/vibrating) extended object can be analyzed by considering the trajectory of its CM.



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Giancoli pg 230 Fig 9.22:

Translational motion of a wrench given by cross and dashed line

Giancoli pg 230 Fig 9.21(b):
Translational Motion of a diver
given by dots and dashed line



(b)
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"All things are difficult before they are easy"

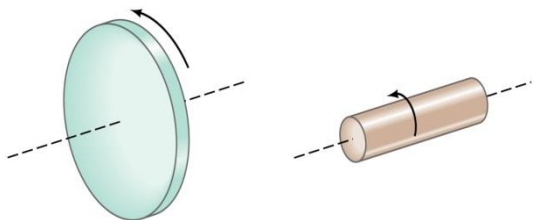
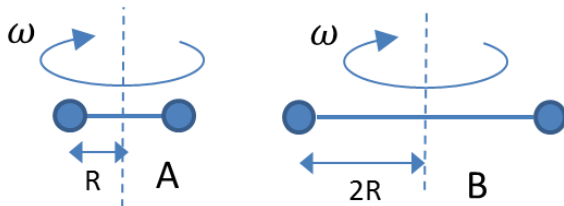
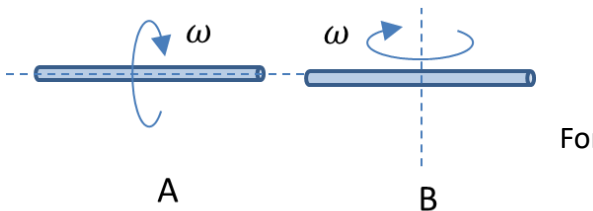
- Thomas Fuller (1608-1661)

"All difficult things have their origin in that which is easy, and great things in that which is small."

- Lao Zi (571? BC -471 BC)

Moment of Inertia

$$I = m_1 r_1^2 + m_2 r_2^2 + \dots + m_n r_n^2 = \sum_{i=1}^n m_i r_i^2$$



$$I = \int R^2 dm$$

Rotational Dynamics

Translational → *Rotational*

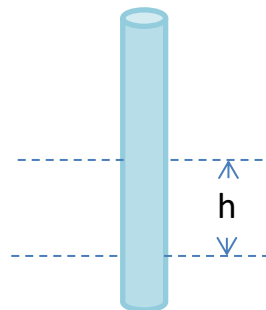
mass m → *Moment of Inertia* I ;

velocity v → *angular velocity* ω ;

acceleration a → *angular acceleration* α

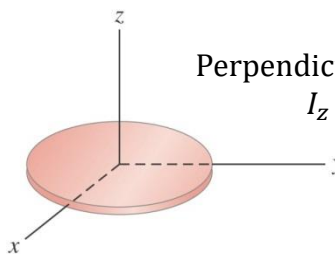
For a fixed axis in inertia reference frame,

$$F = ma \rightarrow \tau = I\alpha ; \text{Kinetic energy} = \frac{1}{2}mv^2 \rightarrow \frac{1}{2}I\omega^2$$



Parallel Axis theorem

$$I = I_{CM} + Mh^2$$

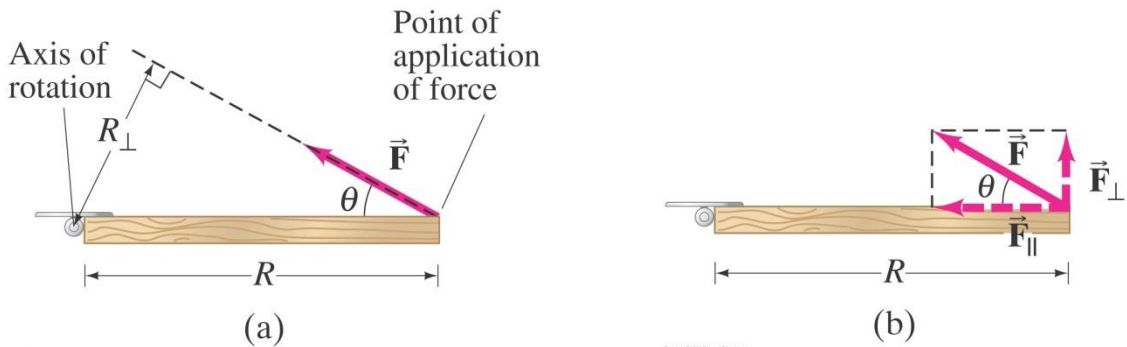


Perpendicular axis theorem

$$I_z = I_x + I_y$$

Definition of Torque (Recap): Giancoli pg 256-7

Torque is the product of force with the perpendicular distance from the axis of rotation to the line which the force acts (lever arm / moment arm).



Giancoli pg 257 Fig 10.14:

Two perspectives of the definition of torque

$$\tau = FR_{\perp} = FR \sin \theta$$

$$\tau = F_{\perp}R = F \sin \theta R$$

Circular Motion (Recap) Giancoli Prob 10.5 (a) A grinding wheel 0.35 m in diameter rotates at 2500 rpm. Calculate its angular velocity in rad/s (b) What are the linear speed and acceleration of a point on the edge of the grinding wheel?

$$a) \omega = 2500 \times \frac{2\pi \text{ rad}}{60 \text{ s}} = 262 \text{ rad/s}$$

$$b) v = r\omega = \frac{0.35 \text{ m}}{2} \times \omega = 45.8 \text{ m/s}$$

$$a = r\omega^2 \left(\text{or } \frac{v^2}{r} \right) = \frac{0.35 \text{ m}}{2} \times \omega^2 = 1.2 \times 10^4 \text{ m/s}^2$$

Circular Motion (Recap), Problem 10.4

The blades of a blender rotate at a rate of 6500 rpm (rev per min). When the motor is turned off during the operation, the blades slowed down to rest in 4.0 s. What is the angular acceleration as the blades slow down?

$$\omega_f = \omega_i + \alpha t$$

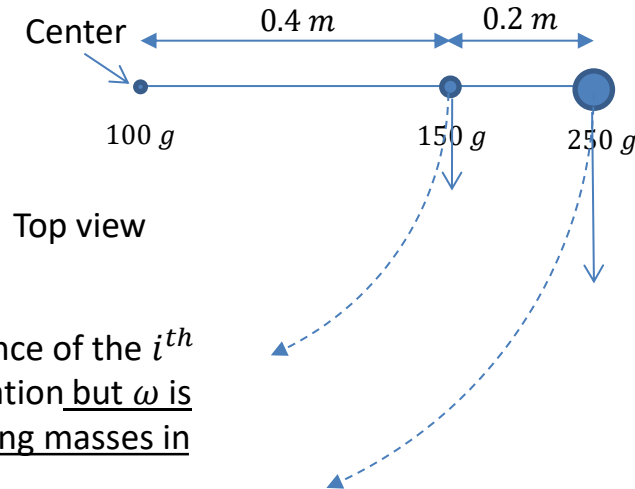
$$0 = 6500 \times \frac{2\pi \text{ rad}}{60 \text{ s}} + \alpha (4.0 \text{ s})$$

$$\alpha = -170 \text{ rad/s}^2$$

Physical Quantities related to Rotation

Recall $s = r\theta$; $v = r\omega$; $a_{tan} = r\alpha$

Consider a system of three point masses connected by a massless rod shown below (top view) and two of the masses are rotating about the center. Mass $m_1 = 150 \text{ g}$ is at a distance $r_1 = 0.4 \text{ m}$ from the center and mass $m_2 = 250 \text{ g}$ is at a distance $r_2 = 0.6 \text{ m}$ from the center. Calculate the kinetic energy of the system if it is rotating about the center at $\omega = 2 \text{ rad/s}$.



v_i changes with distance of the i^{th} mass from axis of rotation but ω is the same for all rotating masses in the system.

$$\begin{aligned}
 \text{Total rotational KE} &= \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 \\
 &= \frac{1}{2}m_1(r_1\omega)^2 + \frac{1}{2}m_2(r_2\omega)^2 \\
 &= \frac{1}{2}(m_1r_1^2 + m_2r_2^2)\omega^2
 \end{aligned}$$

The quantity $(m_1r_1^2 + m_2r_2^2)$ is specific to

- i. the mass distribution of the system and
- ii. the chosen axis of rotation

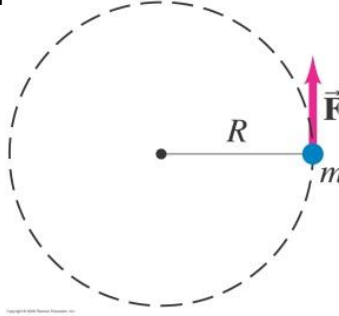
We call this quantity the moment of inertia I of the system (about the chosen axis) and we can write the rotational KE about the chosen axis as

$$KE_{rot} = \frac{1}{2}I\omega^2$$

If we compare with $KE = \frac{1}{2}mv^2$, the moment of inertia I is analogous to mass (in translational motion) and can be a rotational inertia, the reluctance of an object to change its states of rotational motion.

Physical Quantities related to Rotation

We see that the moment of inertia of an object, plays the same role for rotational motion that mass does for translational motion. For simplicity, consider a mass rotating in a circle about a fixed point and a single force F acting at R from the center



The mass start accelerating from rest and using Newton's 2nd law, we write

$$F = ma = mR\alpha \rightarrow FR = \tau = mR^2\alpha = I\alpha$$

Thus, we see from translation dynamics to rotational dynamics.

mass $m \rightarrow$ Moment of Inertia I ;

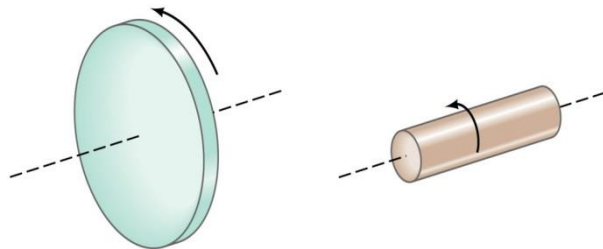
velocity $v \rightarrow$ angular velocity ω ; acceleration $a \rightarrow$ angular acceleration α

For a fixed axis in inertia reference frame,

$$F = ma \rightarrow \tau = I\alpha ; \text{Kinetic energy} = \frac{1}{2}mv^2 \rightarrow \frac{1}{2}I\omega^2$$

And later, *momentum $p = mv \rightarrow$ angular momentum $L = I\omega$*

The same argument can be used for extended objects

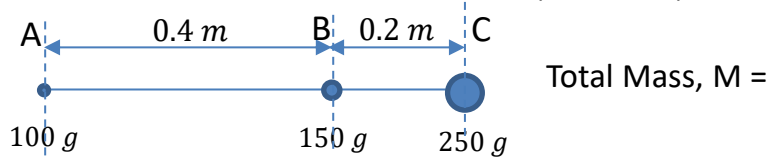


where we write torque τ_i acting on the i^{th} mass can be summed over the whole object $\sum \tau_i = (\sum m_i r_i^2) \alpha$ or

$$\sum \tau = I\alpha \text{ -- always true if axis of rotation is fixed}$$

Determination of moment of inertia of objects for given axis of rotation

Three masses are connected by a rod of negligible mass as shown. Compute the moment of inertia of rod about the axis of rotation (A, B and C) as shown.



Axis of rotation about about A, $I_A =$

$$I_B =$$

$$I_C =$$

Find the center of mass of the system *using A as the reference point*. Find I_{CM} .

$$x_{CM} = \frac{100g \times 0.0m + 150g \times 0.4m + 250g \times 0.6m}{(100g + 150g + 250g)} = 0.42m$$

$$I_{CM} = m_A \times l_{A \rightarrow CM}^2 + m_B \times l_{B \rightarrow CM}^2 + m_C \times l_{C \rightarrow CM}^2$$

$$= (0.10kg)(0.42m)^2 + (0.15kg)(0.02m)^2 + (0.25kg)(0.18m)^2$$

$$= 0.0258kg\ m^2$$

We notice some interesting patterns between the various momentum of inertias I with I_{CM} ...

$$I_{CM} + Ml_{A \rightarrow CM}^2 =$$

$$I_{CM} + Ml_{B \rightarrow CM}^2 =$$

Parallel axis theorem (Giancoli pg 264, see proof on pg 265)

There are two simple theorems useful for obtaining moment of inertia. The first is called the **parallel axis theorem**. It relates the moment of inertia I of an object of total mass M about any axis, and its moment of inertia I_{CM} about an axis passing through the center of mass and parallel to the first axis. If the two axis are a distance h apart, then

$$I = I_{CM} + Mh^2$$

Thus, if the moment of inertia about an axis through the CM is known, the moment of inertia about any parallel axis can be easily obtained.

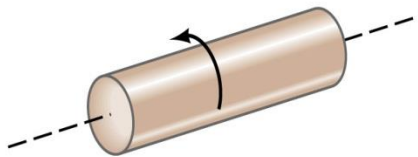
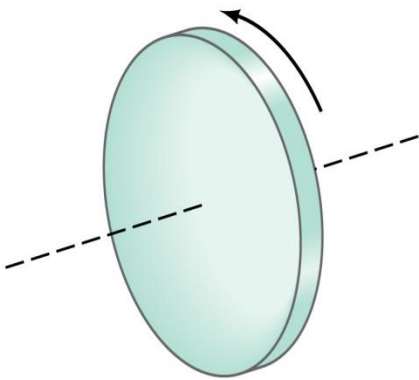
Determination of moment of inertia of objects for given axis of rotation

As we have seen , the quantity

$$I = \sum m_i R_i^2$$

Is the rotational inertia of an object.

The distribution of mass matters here—these two objects have the same mass, but the one on the left has a greater rotational inertia, as so much of its mass is far from the axis of rotation.



<https://www.youtube.com/watch?v=oOtH4Spb9VI&t=5s>

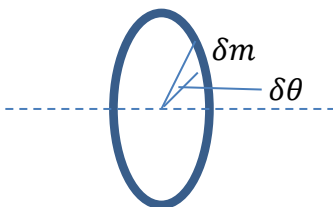


<https://www.youtube.com/watch?v=CHQOctEvtTY&t=7s>

We have dealt with discrete masses previously. Now for an object as shown above, we have to consider them as a continuous distribution of mass, the moment of inertia may be calculated:

$$I = \int R^2 dm$$

As an example, let us compute the moment of inertia of a thin hoop of mass M and radius R about the cylindrical axis. Mass per unit length $\lambda = \frac{M}{2\pi R}$.



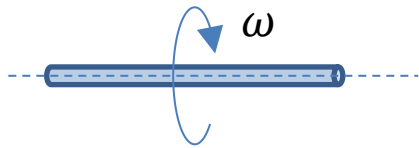
$$\delta I = R^2 dm$$

$$I = \int_0^{2\pi} R^2 \lambda R d\theta = [\lambda R^3]_0^{2\pi} = \lambda R^3 \times 2\pi$$

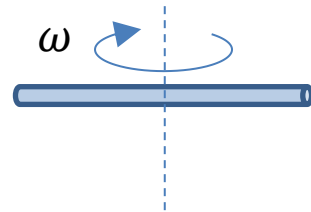
$$= \lambda(2\pi R)R^2 = MR^2$$

Testing your understanding

Consider trying to turn the rod in two different orientations starting from rest and accelerating to angular velocity ω . Which one do you think will require more effort?

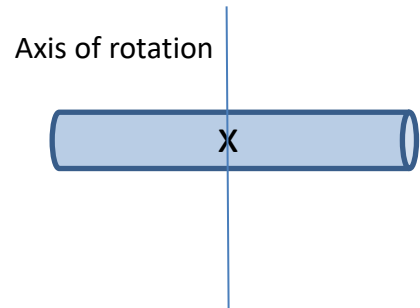
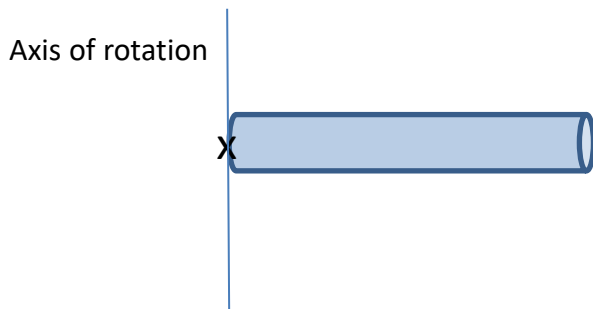


A



B

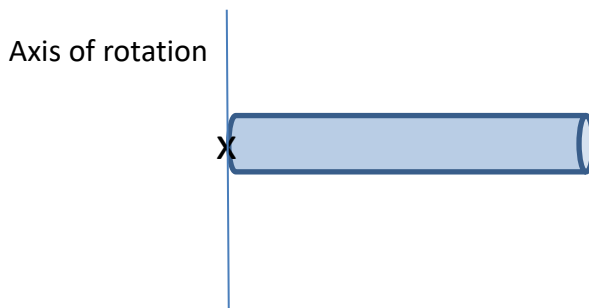
Which has a larger moment of inertia?



Which has a larger moment of inertia?

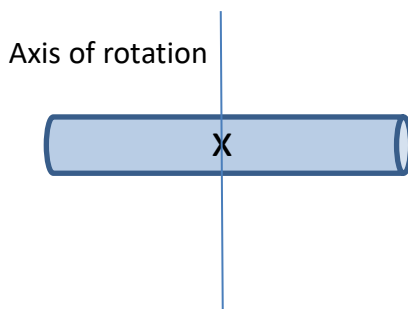
Determination of moment of inertia of objects for given axis of rotation

The mass per unit length λ of a rod is 15 g/cm and its length is $L = 12 \text{ cm}$. Compute the moment of inertia of rod about the axis of rotation as shown.



$$I = \int_0^l$$

Determine the moment of inertia of the same rod when the axis of rotation is about its center of mass.

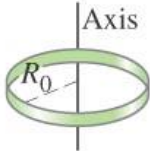
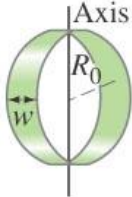
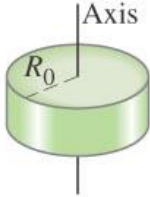
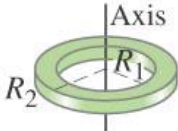
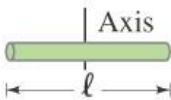
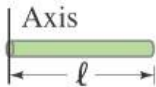
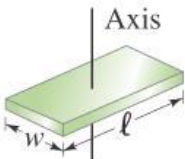


$$I = \int_{-\frac{l}{2}}^{\frac{l}{2}} x^2 dm = \int_{-\frac{l}{2}}^{\frac{l}{2}} x^2 \lambda dx = \frac{\lambda}{3} \left[\left(\frac{l}{2} \right)^3 - \left(-\frac{l}{2} \right)^3 \right] = \frac{\lambda}{3} \frac{2l^3}{8} = \lambda l \times \frac{l^2}{12} = \frac{Ml^2}{12}$$

Check parallel axis theorem:

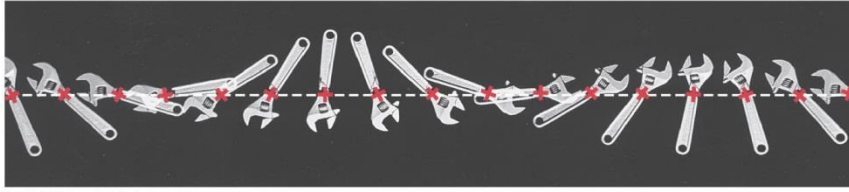
$$I_{\text{end}} = I_{\text{CM}} + M \left(\frac{l}{2} \right)^2$$

Moment of Inertia for various objects with axis of rotation about CM

Object	Location of axis		Moment of inertia
(a) Thin hoop, radius R_0	Through center		MR_0^2
(b) Thin hoop, radius R_0 width w	Through central diameter		$\frac{1}{2}MR_0^2 + \frac{1}{12}Mw^2$
(c) Solid cylinder, radius R_0	Through center		$\frac{1}{2}MR_0^2$
(d) Hollow cylinder, inner radius R_1 outer radius R_2	Through center		$\frac{1}{2}M(R_1^2 + R_2^2)$
(e) Uniform sphere, radius r_0	Through center		$\frac{2}{5}Mr_0^2$
(f) Long uniform rod, length ℓ	Through center		$\frac{1}{12}M\ell^2$
(g) Long uniform rod, length ℓ	Through end		$\frac{1}{3}M\ell^2$
(h) Rectangular thin plate, length ℓ , width w	Through center		$\frac{1}{12}M(\ell^2 + w^2)$

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The importance of Center of mass reference frame



Giancoli pg 230 Fig 9.22:

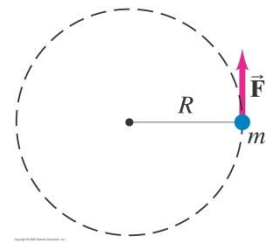
The above diagram shows the motion of a rotating spanner through space with no forces acting on it. We see that if we see that the center of mass is the only point on the spanner that is non-accelerating – making it a special and important reference.

Rotational equivalent for Newton's Second Law

Consider a mass m rotating in a circle of radius R about a fixed point and a single force F acts on m :

$$F = ma = mR\alpha \rightarrow FR = \tau = mR^2\alpha = I\alpha$$

From here, we generalize from a single mass rotating to a rigid extended object with moment of inertia I rotating about a fixed axis.



$$\tau = \sum_i m_i R_i^2 \alpha = I\alpha$$

There can be a few torques acting on the object hence

$$\tau_{net} = \sum \tau = \sum_i m_i R_i^2 \alpha = I\alpha$$

As we have seen in the topic of static equilibrium, torque can be computed from any arbitrary axis and the equation above is only true if the axis is fixed.

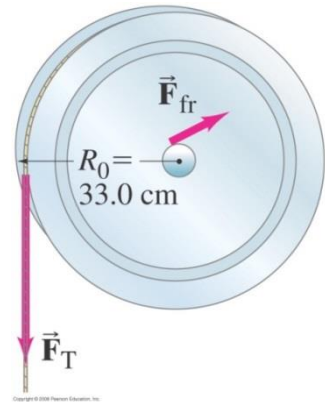
$$\sum \tau = I\alpha - \text{true only for rotation about fixed axis}$$

However, if the torque and moment of inertia I are computed about the center of mass, the relation is also true if the rigid object is translation with acceleration. (e.g. A ball rolling down an inclined plane.)

$$(\sum \tau)_{CM} = I_{CM} \alpha_{CM} - \text{axis direction fixed but may accelerate}$$

Giancoli pg 261, Example 10-9

A 15.0-N force is applied to a cord wrapped around a pulley of mass $M = 4.00 \text{ kg}$ and radius $R_o = 33.0 \text{ cm}$. The pulley accelerates uniformly from rest to an angular speed of 30.0 rad/s in 3.00 s . If there is a frictional torque of $\tau_{fr} = 1.10 \text{ m.N}$ at the axle, determine the moment of inertia of the pulley. The pulley rotates about its center.



$$\alpha = \frac{\Delta\omega}{\Delta t} = 10.0 \text{ rad/s}^2$$

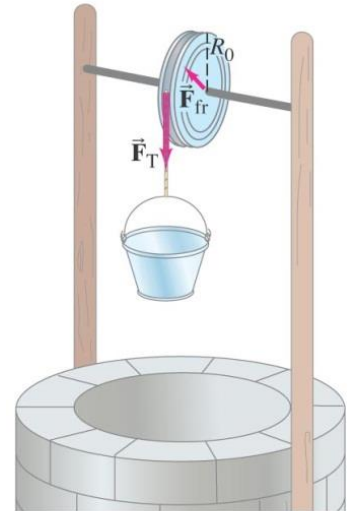
$$\sum\tau = R_o F_T - \tau_{fr} = I\alpha$$

$$\sum\tau = 3.85 \text{ mN}$$

$$I =$$

Giancoli pg 262, Example 10-10

Considering the pulley as before, the cord is now tied to a bucket weight $w = 15.0 \text{ N}$ ($m=1.53 \text{ kg}$). We assume that the cord has negligible mass and does not stretch or slip on the pulley. (a) Calculate the angular acceleration α of the pulley and the linear acceleration of the bucket. (b) Determine the angular velocity ω of the pulley and the linear velocity v of the bucket at $t = 3.00 \text{ s}$ if the pulley (and bucket) start from rest at $t = 0$.



(a)

$$\text{Pulley: } \sum \tau = R_o F_T - \tau_{fr} = I \alpha \quad (1)$$

$$\text{Bucket: } \sum F = mg - F_T = ma \Rightarrow F_T = mg - ma \quad (2)$$

$$\text{Rope does not slip: } a_{tan} = R_o \alpha (= a) \quad (3)$$

$$I \alpha = R_o (mg - m(R_o \alpha)) - \tau_{fr}$$



(b)

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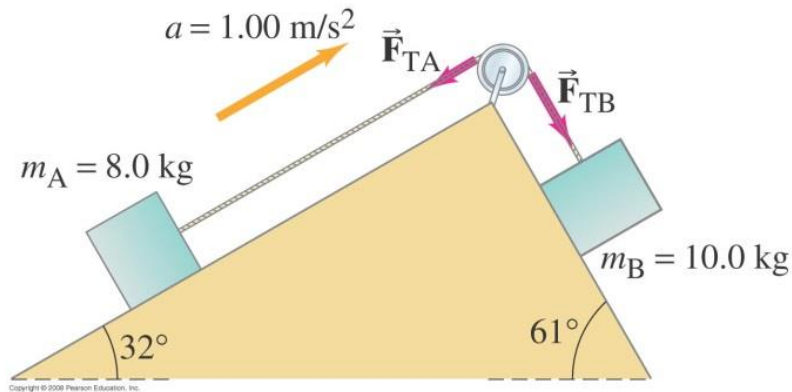
$$\alpha =$$

$$(b) \omega = \omega_o + \alpha t =$$

$$v = r\omega =$$

Giancoli Prob 10.46

Two blocks are connected by a light string passing over a pulley of radius 0.15 m and moment of inertia I . The blocks move (towards the right) with an acceleration of a along their frictionless inclines (see Fig. 10–54). (a) Draw free-body diagrams for each of the two blocks and the pulley. (b) Determine a and the tensions in the two parts of the string. (c) Find the net torque acting on the pulley, and determine its moment of inertia, I .



(b) m_A along slope: $F_{TA} - m_A g \sin \theta_A = m_A a$

$$F_{TA} =$$

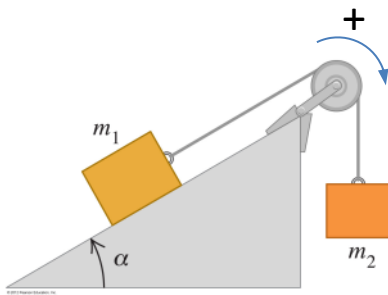
m_B along slope: $m_B g \sin \theta_B - F_{TB} = m_B a$

$$F_{TB} =$$

(c) Pulley:

$$I = 0.59 \text{ kg m}^2$$

Progression: Blocks and Pulley (Forces)

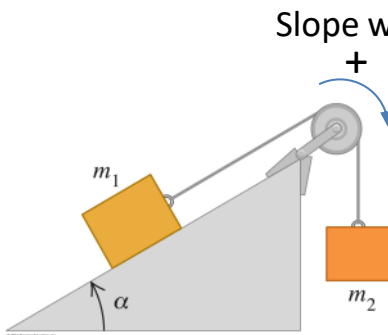


+ Frictionless slope, smooth pulley
Forces

$$m_2 g - T = m_2 a$$

$$F_n - m_1 g \cos \alpha = 0$$

$$T - m_1 g \sin \alpha = m_1 a$$



Slope with Friction, μ_k ; smooth pulley; m_1 sliding up

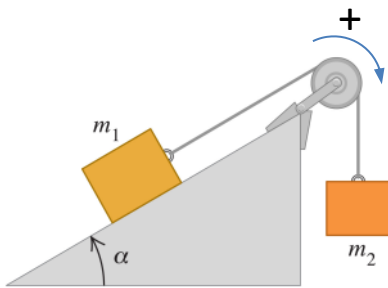
+
Forces

$$m_2 g - T = m_2 a$$

$$F_n - m_1 g \cos \alpha = 0$$

$$T - m_1 g \sin \alpha - \mu_k m_1 g \cos \alpha = m_1 a$$

Slope with Friction, pulley rotates without slipping; m_1 sliding up



+
Forces

$$m_2 g - T_2 = m_2 a$$

$$F_n - m_1 g \cos \alpha = 0$$

$$T_1 - m_1 g \sin \alpha - \mu_k m_1 g \cos \alpha = m_1 a$$

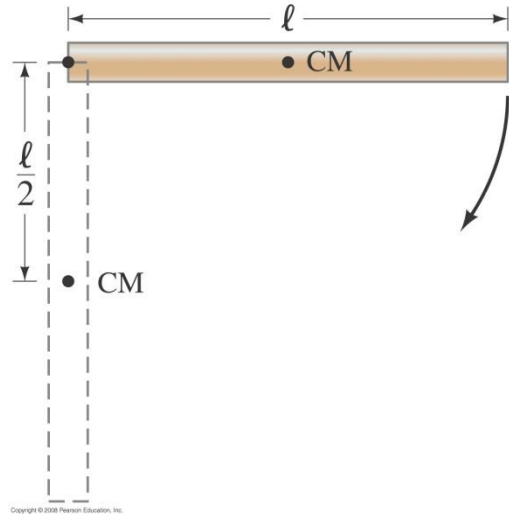
$$(T_2 - T_1) R_{\text{pulley}} = I_{\text{pulley}} \alpha$$

$$\text{No slipping: } a = R_{\text{pulley}} \alpha$$

Giancoli pg 263, 267, Example 10-11/10-15

A uniform rod of mass M and length l is pivoted on a frictionless hinge as shown. The rod is held horizontally and then released. At the moment of release, determine

- (a) the angular acceleration of the rod,
 - (b) the linear acceleration at the tip of the rod,
 - (c) the angular velocity of the rod when it reaches the vertical position, and
 - (d) the speed of the rod's tip at this moment.
- (assume weight of rod acts through CM)



a) Acceleration => consider

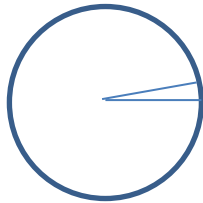
c) Velocity => consider



<https://www.youtube.com/watch?v=eVBP1sz-z74>

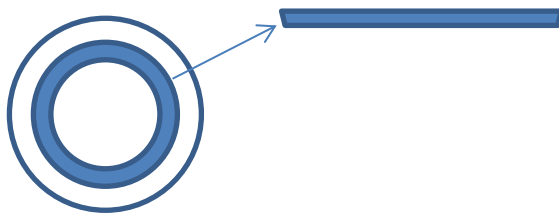
Appendix 1: A Mathematical Digression (on integration) [Non-examinable]

1. Deriving *circumference* = $2\pi R$



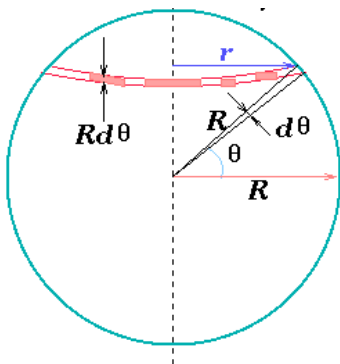
$$C = \int_0^{2\pi} R d\theta = (2\pi - 0)R = 2\pi R$$

2. Deriving Area of circle = πR^2



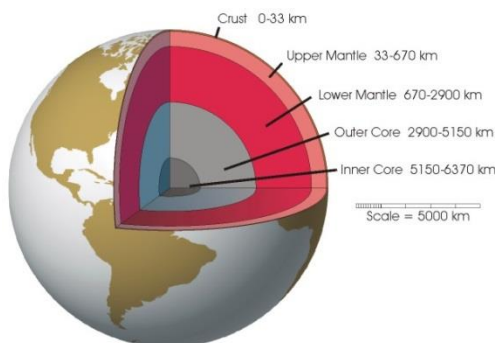
$$\begin{aligned} A &= \int_0^R 2\pi r dr = 2\pi \int_0^R r dr \\ &= 2\pi \left(\frac{r^2}{2} - \frac{0^2}{2} \right) = \pi R^2 \end{aligned}$$

3. Deriving Surface Area of sphere = $4\pi R^2$



$$\begin{aligned} S &= \int_{-\pi}^{\pi} 2\pi r R d\theta = 2\pi R \int_{-\pi/2}^{\pi/2} R \cos \theta d\theta \\ &= 2\pi R^2 \left(\sin \frac{\pi}{2} - \sin \left(-\frac{\pi}{2} \right) \right) = 4\pi R^2 \end{aligned}$$

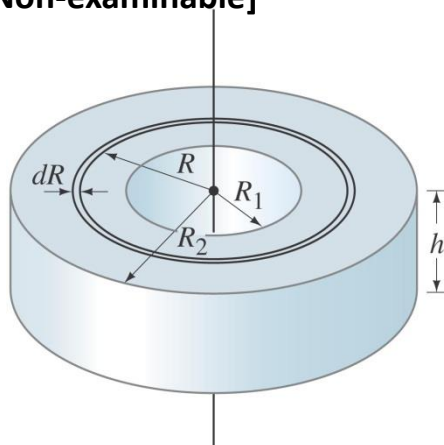
4. Deriving Volume of sphere = $\frac{4}{3}\pi R^3$



$$\begin{aligned} S &= \int_0^R 4\pi r^2 dr = 4\pi \int_0^R r^2 dr \\ &= 4\pi \left(\frac{R^3}{3} - \frac{0^3}{3} \right) = \frac{4}{3}\pi R^3 \end{aligned}$$

Appendix 1: Giancoli pg 264, Example 10-12[Non-examinable]

- (a) Show that the moment of inertia of a uniform hollow cylinder of inner radius R_1 , outer radius R_2 , and mass M , is $I = \frac{1}{2} M(R_1^2 + R_2^2)$, if the rotation axis is through the center along the axis of symmetry.
- (b) Obtain the moment of inertia for a solid cylinder.



APPROACH We know that the moment of inertia of a thin ring of radius R is mR^2 . So we divide the cylinder into thin concentric cylindrical rings or hoops of thickness dR , one of which is indicated in Fig. 10-24. If the density (mass per unit volume) is ρ , then

$$dm = \rho dV,$$

where dV is the volume of the thin ring of radius R , thickness dR , and height h . Since $dV = (2\pi R)(dR)(h)$, we have

$$dm = 2\pi\rho h R dR.$$

SOLUTION (a) The moment of inertia is obtained by integrating (summing) over all these rings:

$$I = \int R^2 dm = \int_{R_1}^{R_2} 2\pi\rho h R^3 dR = 2\pi\rho h \left[\frac{R_2^4 - R_1^4}{4} \right] = \frac{\pi\rho h}{2} (R_2^4 - R_1^4),$$

where we are given that the cylinder has uniform density, $\rho = \text{constant}$. (If this were not so, we would have to know ρ as a function of R before the integration could be carried out.) The volume V of this hollow cylinder is $V = (\pi R_2^2 - \pi R_1^2)h$, so its mass M is

$$M = \rho V = \rho\pi(R_2^2 - R_1^2)h.$$

Since $(R_2^4 - R_1^4) = (R_2^2 - R_1^2)(R_2^2 + R_1^2)$, we have

$$I = \frac{\pi\rho h}{2} (R_2^2 - R_1^2)(R_2^2 + R_1^2) = \frac{1}{2} M(R_1^2 + R_2^2),$$

as stated in Fig. 10-20d.

(b) For a solid cylinder, $R_1 = 0$ and if we set $R_2 = R_0$, then

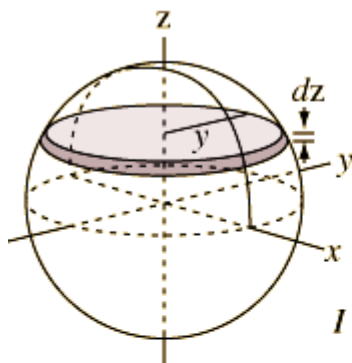
$$I = \frac{1}{2} MR_0^2,$$

which is that given in Fig. 10-20c for a solid cylinder of mass M and radius R_0 .

Appendix 1: Moment of Inertia of a Sphere (using MI of discs)

[Non-examinable]

<http://hyperphysics.phy-astr.gsu.edu/hbase/isph.html>



$$dI = \frac{1}{2} y^2 dm = \frac{1}{2} y^2 \rho dV = \frac{1}{2} y^2 \rho \pi y^2 dz$$

and the integral becomes

$$I = \frac{1}{2} \rho \pi \int_{-R}^R y^4 dz = \frac{1}{2} \rho \pi \int_{-R}^R (R^2 - z^2)^2 dz = \frac{8}{15} \rho \pi R^5$$

Radius = R

Mass = M

$$\text{Density} = \rho = \frac{M}{V} = \frac{M}{\frac{4}{3} \pi R^3}$$

Substituting the density expression gives

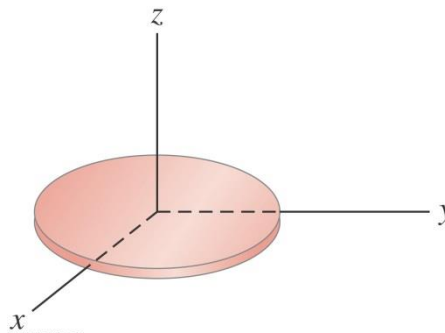
$$I = \frac{8}{15} \left[\frac{M}{\frac{4}{3} \pi R^3} \right] \pi R^5 = \frac{2}{5} MR^2$$

Perpendicular Axis Theorem [Non-examinable]

The parallel axis theorem that we have seen earlier can be applied to any object. However, the perpendicular-axis theorem can only be applied to plane (flat) objects – objects which thickness can be neglected compared to other dimensions. The theorem states that:

The sum of moment of inertia of a plane object about any two perpendicular axes in the plane of the object, is equal to the moment of inertia about an axis through their point of intersection perpendicular to the plane of the object. For an object in the xy plane

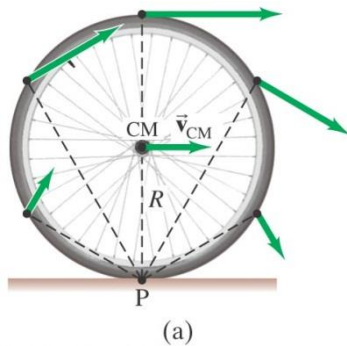
$$I_z = I_x + I_y$$



Here, I_z, I_x, I_y are the moment of inertia about the z, x and y axes. The proof is simple: since $I_x = \sum m_i y_i^2$, $I_y = \sum m_i x_i^2$, and $I_z = \sum m_i (x_i^2 + y_i^2)$.

"Success is the ability to go from one failure to another with no loss of enthusiasm."
- Winston Churchill (1874-1965)

Rolling (without slipping)

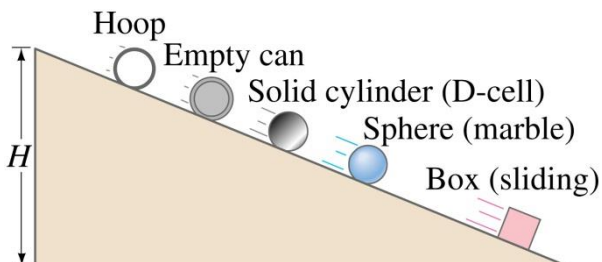


Giancoli pg 268 Fig 10.31

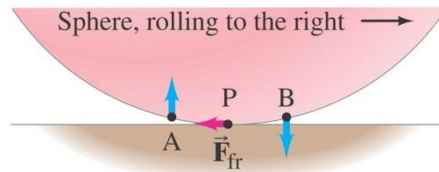
$$v = R\omega$$

$$\sum \tau_{CM} = \frac{dL_{CM}}{dt} = I_{CM}\alpha_{CM}$$

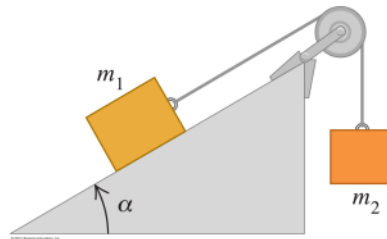
$$K_{tot} = \frac{1}{2} I_{CM}\omega^2 + \frac{1}{2} Mv_{CM}^2$$



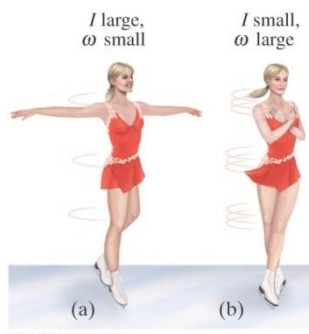
Giancoli pg 269 Fig 10.34



Giancoli pg 270 Fig 10.35

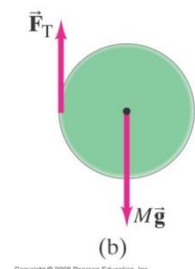


Pulley: $\sum \tau = I\alpha$;
No slipping : $a = R\alpha$



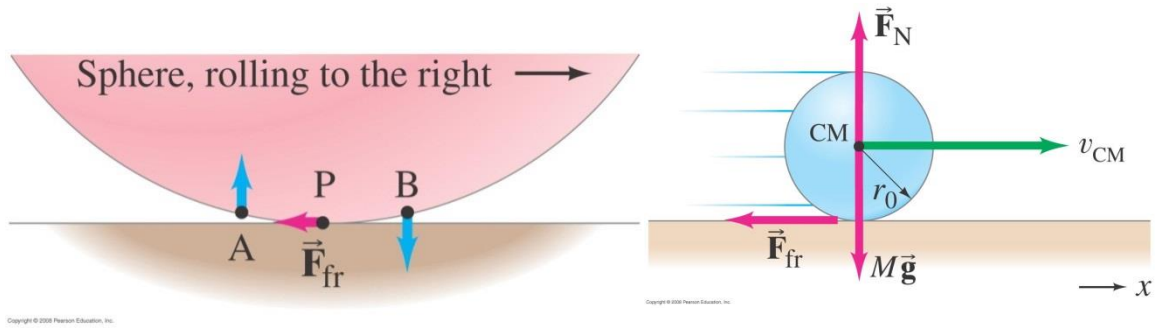
$$I\omega = I\omega_o$$

Newton's second law (rotation)



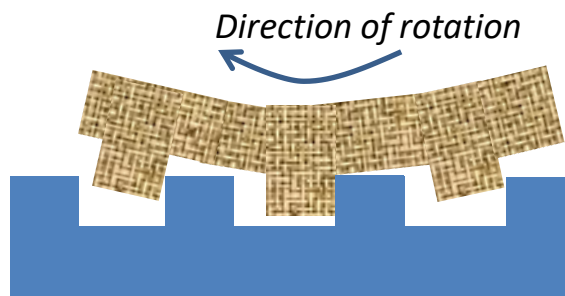
Giancoli pg 271 Fig 10.73

Sufficient Friction is needed for Round objects to roll: Giancoli pg 270

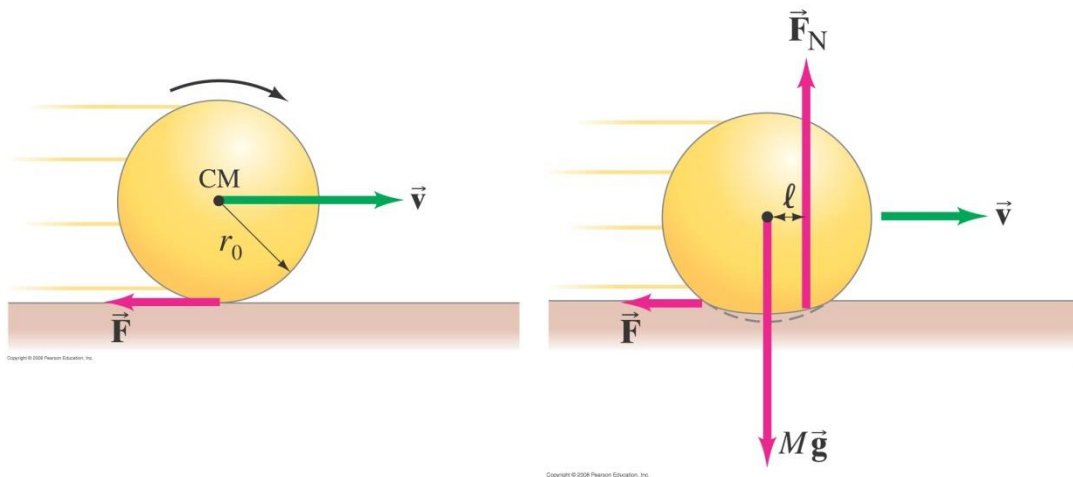


Giancoli pg 270 Fig 10.35, 10.38:

Without static friction, round objects would slide rather than roll. If there is not sufficient static friction, the object will roll and slide at the same time.



Why does a rolling sphere slow down?: Giancoli pg 273-274



Giancoli pg 273 Fig 10.40-41:

Why does a rolling sphere slow down?

No real sphere is perfectly rigid. The bottom will deform, and the normal force will create a torque that slows the sphere.

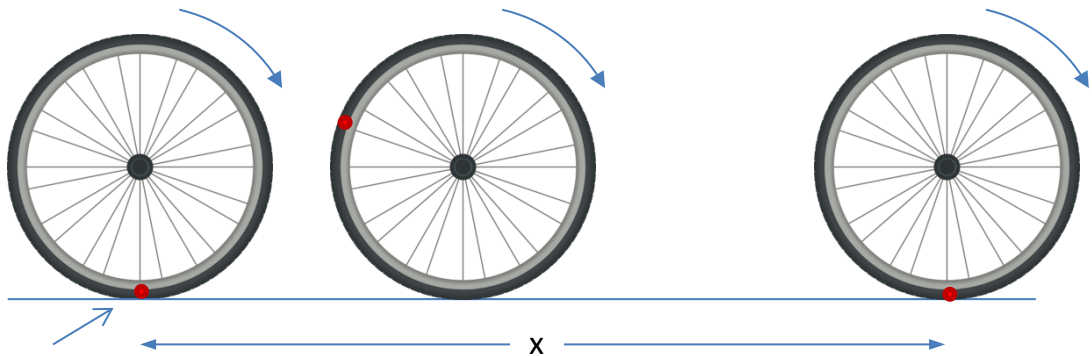
Rolling Friction

Rolling Friction – easier to move a sack of rice across floor using a trolley with wheels than to slide it. To compare the frictional forces involve in both cases, we define a **coefficient of rolling friction** μ_r , which is the frictional force needed for constant speed on flat surface divided by the normal force exerted by the surface.

Typical values of μ_r are 0.002 to 0.003 for steel wheels on steel rails and 0.01 to 0.02 for rubber tyres on concrete. This is one reason why railtrains are more fuel efficient compared to highway trucks.

Rolling without slipping

A wheel of radius R is rolling along a horizontal road (without slipping) at constant angular velocity ω . A red dot has been marked on the rim of the wheel.



What is the angular displacement of the wheel when it completes one round? 2π

What is the distance travelled by the red dot when it completes one round? $2\pi R$

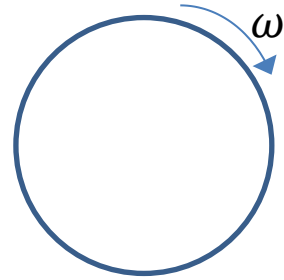
When the red dot completes one round, what is the distance x travelled by the wheel (center of the wheel)? $x = 2\pi R$

If the wheel (or red dot) takes time T to complete one round, what is the velocity of center of mass of the wheel? $v = \frac{x}{T} = \frac{2\pi R}{T} = \omega R$ (when there is no slipping)

Rotational Kinetic Energy

For a disc rotating about the axis passing through its center of mass, the rotational kinetic energy is given by

$$K.E._{rot} = \frac{1}{2} \sum m_i r_i^2 \omega^2 = \frac{1}{2} I \omega^2$$



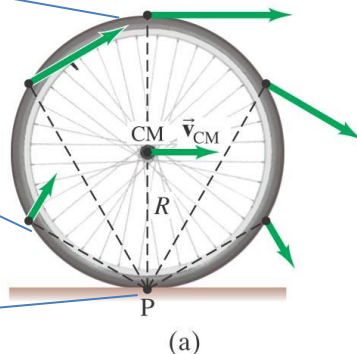
Giancoli pg 268 All about Rolling without slipping

When the wheel rolls without slipping, the point of contact of the wheel with the ground is instantaneously at rest.

Points further from ground have greater linear speed

Points closer to ground have small linear speed

“Instantaneous axis” passing point P



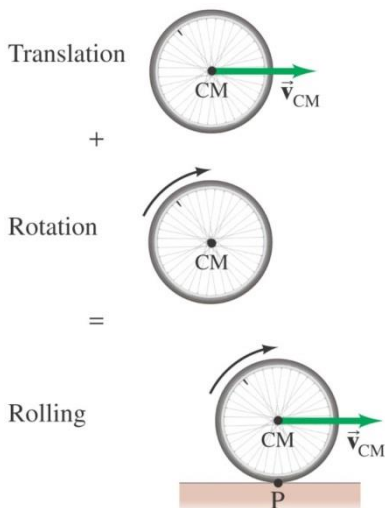
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In the photo, spokes nearer top appear blurry because they are moving faster



(b)

Giancoli pg 268 Fig 10.31



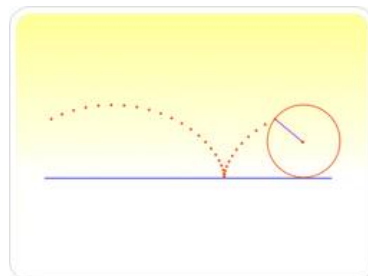
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Giancoli pg 268 Fig 10.32

$$K_{tot} = \frac{1}{2} I_{CM} \omega^2 + \frac{1}{2} M R^2 \omega^2$$

$$K_{tot} = \frac{1}{2} I_{CM} \omega^2 + \frac{1}{2} M v_{CM}^2$$

$$K_{tot} = K_{rot} + K_{trans}$$



Giancoli pg 268 Example 10-16

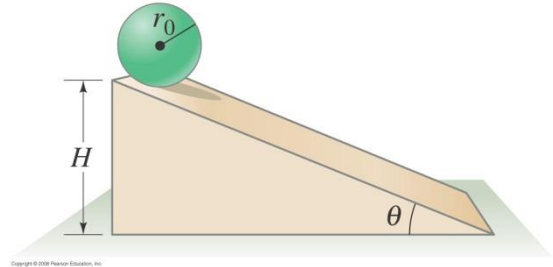
What will be the speed of a solid sphere of mass M and radius r_0 when it reaches the bottom of an incline if it starts from rest at a vertical height H and rolls without slipping? (Assume no slipping occurs because of static friction, which does no work.) Compare your result to that for an object sliding down a frictionless incline.

Total energy at any vertical distance y above base of incline is

$$\frac{1}{2}Mv^2 + \frac{1}{2}I_{CM}\omega^2 + Mgy$$

By applying conservation of energy,

Total energy on top of incline = total energy at the bottom of incline



Checking the tables, $I_{CM} = \frac{2}{5}Mr_0^2$,



<https://www.youtube.com/watch?v=cB8GNQuyMPc&t=5s>

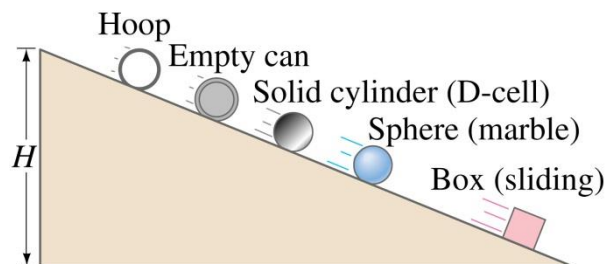
Giancoli pg 267, Example 10-17

Several objects roll without slipping down an incline of vertical height H , all starting from rest at the same moment. In what order do they reach the bottom of the incline?

$$I_{Hoop, CM} = MR^2$$

$$I_{D-Cell, CM} = \frac{1}{2}MR^2$$

$$I_{Marble, CM} = \frac{2}{5}MR^2$$



The speed at the bottom does not depend on the object's mass or radius but only on shape and height of the hill.

Giancoli pg 271, Example 10-19

A falling yo-yo – String is wrapped round a uniform solid cylinder of mass M and radius R , and the cylinder starts falling from rest. As the cylinder falls, find (a) its acceleration and (b) the tension in the string.

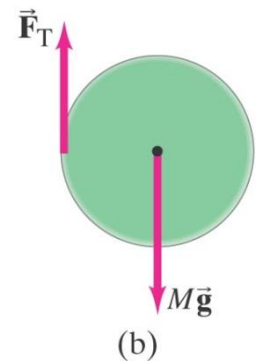
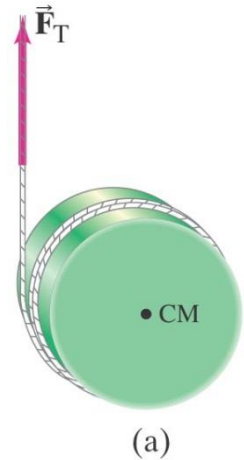
Newton's Law for translational motion

$$\begin{aligned}\sum F &= Ma; \\ Mg - F_T &= Ma\end{aligned}$$

Newton's Law for rotational motion

$$\begin{aligned}\sum \tau_{CM} &= I_{CM}\alpha_{CM} \\ F_T R &= \frac{1}{2}MR^2\alpha\end{aligned}$$

No slipping: $a = \alpha R$



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Angular Momentum and Torque [Non-examinable] MAE, CE – Video after Exam

Considering an object with moment of inertia I rotating about an axis through the center of mass with angular velocity ω , we can write the angular momentum L as

$$L = I\omega$$

Applying the rotational equivalent of Newton's second law, we can write

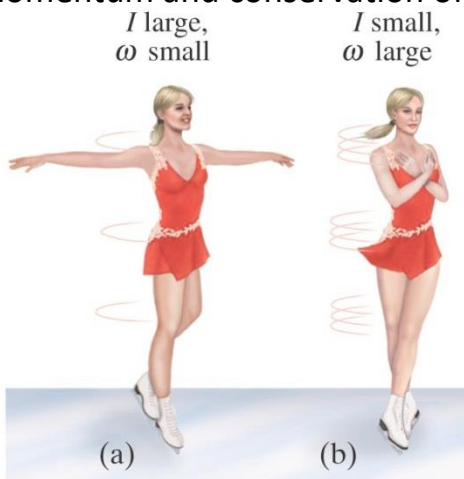
$$\sum \tau = \frac{dL}{dt} = I\alpha,$$

for constant I . *This above relation is valid for rotational motion about fixed axis and for moving object if its rotation is about an axis passing through the center of mass.*

If the net external torque $\sum \tau = 0$, then

$$\frac{dL}{dt} = 0 \Rightarrow L = I\omega = \text{constant}.$$

This is the law of conservation of angular momentum – one of the great conservation laws in Physics, together with conservation of linear momentum and conservation of energy.



Giancoli pg 286 Fig 11.

$$I\omega = I\omega_0$$



<https://www.youtube.com/watch?v=AQLtcEAG9v0>

Example for Conservation of Angular Momentum [Non-examinable]
Giancoli pg 304 Problem 11-10 MAE, CE – Video after Exam

A uniform disk turns at 3.7 rev/s around a frictionless spindle. A nonrotating rod, of the same mass as the disk and length equal to the disc's diameter, is dropped onto the freely spinning disk as shown in the figure. They then turn together around the spindle with their centers superposed. What is the angular frequency in of the combination?

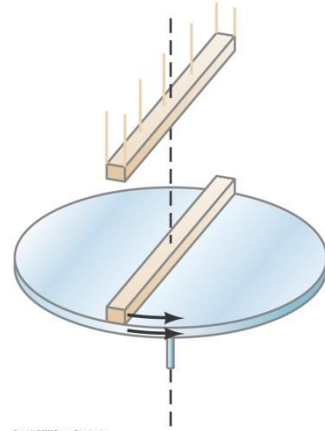
By conservation of angular momentum,

$$\sum L_i = \sum L_f$$

$$I_{rod} \times 0 + I_{disc} \times \omega_i = (I_{rod} + I_{disc})\omega_f$$

$$\omega_f = \frac{I_{disc}\omega_i}{(I_{rod} + I_{disc})} =$$

$$\left[\frac{\frac{1}{2}MR^2}{\frac{1}{2}MR^2 + \frac{1}{12}M(2R)^2} \right] \omega_i = \frac{3}{5} (3.7 \text{ rev/s}) =$$



Giancoli pg 298, Example 11-12

A bullet of mass m moving with velocity v strikes and becomes embedded at the edge of a cylinder of mass M and radius R_0 . The cylinder, initially at rest, begins to rotate about its symmetry axis, which remains fixed in position. Assuming no frictional torque, what is the angular velocity of the cylinder after this collision?

Total angular momentum of bullet about O
 (before collision):

$$L_{bef} = |\vec{r} \times \vec{p}| = R_0 mv$$

Total angular momentum of system about O
 (after collision):

$$L_{aft} = I\omega = (I_{cyl} + mR_0^2)\omega = \frac{1}{2}MR_0^2\omega + mR_0^2\omega$$

Conservation of angular momentum

$$R_0 mv = \frac{1}{2}MR_0^2\omega + mR_0^2\omega$$

$$\omega = \frac{mv}{\left(\frac{1}{2}M + m\right)R_0}$$

